**Old age people are sick often, why?**
Due to degeneration of thymus gland, thymosine level decreases, as a result the immunity of old age people becomes weak and causes sickness.

### 11.2.7 Adrenal gland

A pair of adrenal glands are located at the anterior end of the kidneys, hence also called suprarenal glands. Anatomically the outer region is the cortex and the inner region is the medulla. Histologically the adrenal cortex has three distinct zones, zona glomerulosa, zona fasciculata and zona reticularis. **Zona glomerulosa** an outer thin layer constitutes about 15% of adrenal cortex, and secretes mineralocorticoids. **Zona fasciculata**, the middle widest layer constitutes about 75% of adrenal cortex and secretes glucocorticoids such as cortisol, corticosterone and trace amounts of adrenal androgen and oestrogen. **Zona reticularis**, an inner zone of adrenal cortex constitute about 10% of adrenal cortex and secretes the adrenal androgen, trace amount of oestrogen and glucocorticoids.

Laughing is good for health, because it reduces the stress hormone (adrenalin) secretion and makes us to relax.

**Adrenal medulla:** It is the central part of adrenal gland and is composed of ovoid and columnar cells, which are found around the network of blood capillaries. Adrenalin (epinephrine) and nor adrenalin (nor epinephrine) are the two hormones secreted by the adrenal medulla. Both adrenalin and nor adrenalin are **catecholamines.**

**Function of adrenal hormones**: **Glucocorticoids** stimulate gluconeogensis, lipolysis and proteolysis (the life saving activity). **Cortisol** is a glucocorticoid involved in maintaining cardio vascular and kidney functions. It produces anti-inflammatory reactions and suppresses the immune response. It stimulates the RBC production. It is also known as stress combat hormone. **Mineralocorticoid**s regulates water and electrolyte balance of our body. **Aldosterone** stimulates the reabsorption of sodium and water and eliminates potassium and phosphate ions through excretion, thus it helps in maintaining electrolytes, osmotic pressure and blood pressure. Adrenal androgen plays a role in hair growth in the axial region, pubis and face during puberty.

The **adrenal medulla** secretes the hormones adrenalin and noradrenalin and are referred as "3F hormone" (fight, flight and fright hormone). Adrenalin increases liver glycogen breakdown into glucose and increases the release of fatty acids from fat cells. During emergency it increases heart beat rate and blood pressure. It stimulates the smooth muscles of cutaneous and visceral arteries to decrease blood flow. It increases blood flow to the skeletal muscles thereby increases the metabolic rate of skeletal muscles, cardiac muscles and nervous tissue.

The general function of noradrenalin is to mobilize the brain and body for action. It's secretion is less during sleep, more during wakefulness and reaches much higher levels during stress situations. This response is known as **'fight or flight'** response.



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